# HTR with a local LLM

Use a locally running LLM to perform HTR and similar tasks. Nothing leaves your computer.

Workshop, path 1: LM Studio

# What is in the folder

Download `letters.zip` from the workshop's repository and unzip it.

The folder `letters` holds three letter pages, numbered 1 to 3, each as a picture (`.jpg`) and as a transcription (`.txt`).

It also holds `cer.html`, a page that computes the character error rate (CER). Double-click it to open it in your browser; it works offline.

# 0 Does it fit?

Find out how much memory your laptop has.

Mac: Apple menu → *About This Mac*, the line *Memory*

Windows: *Settings* → *System* → *About*, the line *Installed RAM*

LM Studio needs a Mac with Apple Silicon and macOS 14 or newer, or a Windows laptop.

With an Intel Mac or less than 8 GB, pair up with someone whose laptop fits, or work on the notebooks instead.

# 0 Five sizes of Gemma 4

| Gemma 4 | memory | download size Mac | download size Windows |
|---|---|---|---|
| E2B | 8 GB | 4.3 GB | 4.4 GB |
| E4B | 16 GB | 6.8 GB | 6.3 GB |
| 12B | 16 GB | 6.7 GB | 7.6 GB |
| 26B A4B | 32 GB | 15.6 GB | 18.0 GB |
| 31B | 32 GB | 18.4 GB | 19.9 GB |




Each model needs a laptop with at least the memory shown.

# 0 Which one?

Take the largest model your memory allows.

The E2B is the fastest but makes the most errors; it is the one for 8 GB.

The E4B is the usual choice with 16 GB. The 12B is a little more exact if you close other programs.

With 32 GB, the 26B A4B is nearly as exact as the 31B and much faster.

The 31B is the most exact and the slowest.

# 0 What the names mean

B stands for billions of parameters, the numbers a model computes with; the lecture's filter had nine. In the 12B and the 31B, all of them work on every word.

E stands for *effective*. The E2B and E4B work with 2.3 and 4.5 billion parameters, but have 5.1 and 8 billion with their lookup tables, so the download is larger than the name suggests.

A stands for *active*. In the 26B A4B, 8 of 128 experts work on each word, 3.8 billion parameters: it needs the memory of a 26B but runs nearly as fast as a 4B.

# 0 Mac: MLX. Windows: GGUF.

On a Mac, take *MLX*, Apple's own framework for its chips; it comes with LM Studio.

On Windows, take *GGUF*.

Take the *4-bit* version (in GGUF: Q4_K_M). It stores each number in 4 bits instead of 16, so it is smaller and a little less exact.

If a model will not load, update LM Studio.

# 0 Install

Install LM Studio from lmstudio.ai.

In *Model Search*, search for `gemma-4` and download the model you chose (Mac: MLX, 4-bit; Windows: GGUF).

In the *Chat* tab, load the model at the top and ask it a question.

# 1 Context length




When you load the model, LM Studio asks for a context length.

The context is what the model can hold at once: your prompt, the picture and its own answers so far, counted in tokens, pieces of words.

The default of 8 192 holds a letter page, its picture and the answer with room to spare.

# 1 When the context is full

Everything in a chat stays in the context, so it fills as you go on.

When it is full, LM Studio quietly drops the middle of the conversation (the setting *Context Overflow*, *Truncate Middle*).

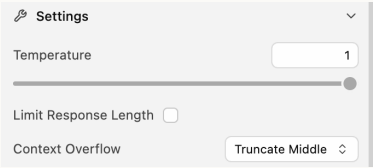
That is why each attempt in this guide is a new chat.

If you want several pages in one chat, load the model again with 16 384 or 32 768. A larger context takes more memory.

# 1 Temperature

Open the panel on the right of the chat. Under *Settings* you find the temperature.

It sets how freely the model picks each next word. At 0 it takes the most likely word every time, so the same page gives the same text. At 1, the default, the text varies from run to run.



# 1 Which temperature?

Set 0 for reading a page. The thesis behind yesterday's lecture read every page at 0, and a rerun gave the same text 57 times out of 57.

For a summary or a translation either is fine.

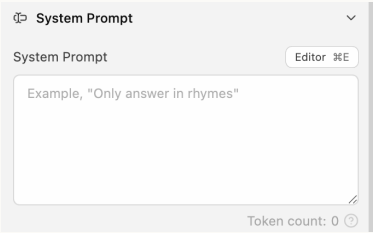
Once your prompt works, run the same page at 1 and compare. What changes?

# 1 The system prompt

At the top of the same panel is the system prompt: text that is always added to your prompt, your defaults.

Once your prompt works, put it there. Every new chat starts with it, and your message is only the picture.

Leave the other settings as they are.



# 2 First reading

Start a new chat and attach letter 1 (`1-joseph-1868-03-27.jpg`).

Ask, in your own words, for the text of the page.

Compare the reply with the scan.

# 3 Measure

Open `cer.html`.

Paste letter 1's transcription (the `.txt`) into *Reference* and the model's text into *Model output*.

The page shows the character error rate (CER) and marks every difference.

# 4 Refine the prompt

Look at what the model got wrong and add an instruction to your prompt that would have prevented it.

Ask again with the new prompt in a new chat, so that the model does not see its first answer, and measure again.

Repeat this two or three times.

# 5 Another letter

Try your best prompt, unchanged, on letter 2 and then on letter 3, each in a new chat, and measure both.

Does the prompt still work, or did it only suit letter 1?

Change it until one prompt works on all three letters, and measure all three again.

# 6 Beyond transcription

Ask for something other than the text, on any of the letters.

For example, ask for a list of the people, places and dates, and check each against the transcription. What did the model invent?

Or ask for a summary in two sentences, or a Swedish translation. What is missing, and what was added?

# 7 Your own material

Take a page of your own and try your best prompt on it.

What is different about this page, and did the model notice?

# 7 What to tell the model

Say what the page is, in which language and from which century.

Ask it to keep the writer’s spelling and line breaks, and to mark what it cannot read with [?] instead of filling the gap.

Names, dates and amounts are where it invents. Check each against the page.

# 7 Letters and diaries

Ask for the text first, one page per chat, at temperature 0.

Then ask for the people, places and dates, or for a summary, and check each against the page.

# 7 Maps

Ask for the place names it can read, as a list. Then ask for the legend, the scale, the region and the date it would guess, and what it is unsure of.

Check the names against the map. It guesses region and date from the style and is confident when wrong.